13.7 Exploring Transformations by a Factor

Lesson Introduction

Benchmark	MA.912.F.2.1 Identify the effect on the graph or table of a given function after replacing $f(x)$ by $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k.	Learning Targets	• I can discover the effect on the graph of $f(x)$ for a transformation of $f(kx)$ or $kf(x)$. • I can discover the effect on the table of values for $f(x)$ for a transformation of $f(kx)$ or $kf(x)$.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How can I determine the effect on the graph of $f(x)$ for a transformation of $f(kx)$ *or* $kf(x)$? • How can I determine the effect on a table of values of $f(x)$ for a transformation of $f(kx)$ *or* $kf(x)$?
Lesson Narrative	In this lesson, students continue their work with transformations, moving to exploring transformations by a factor. Students will begin by exploring transformations that aren't shifted but rather are transformed by a factor. They will bring together what they did during the Exploration activity to learn about the effect of transformations of factors. The lesson will conclude with practice with transformations by a factor using functions of different types.

Component Summary				
	13.7.1 Warm-Up (~5 min.)	Notice and Wonder pattern activity *(SE p. 268)*	13.7.3 Bring It Together (10-15 min.)	The effect of transformations table activity *(SE p. 271)*
	13.7.2 Exploration (10-15 min.)	Transformations by a factor exploration activity *(SE p. 268)*	13.7.4 Guided Practice (10-15 min.)	Transformations practice activity *(SE p. 272)*
	13.7.5 Wrap-Up (~5 min.)	Ticket Out the Door on transformations by a factor *(SE p. 272)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• Desmos links for parent functions	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may mix up the impact of factors with different number sizes.
Students may misinterpret the transformation when the factor is a negative number.</td><td>In 13.7.2 Exploration, students are exploring the transformation of functions by a given factor. There are points where students may get confused and struggle. Be cautious not to give away anything during the Exploration, but instead, allow the Exploration to unfold naturally as designed.
Consider using the Desmos links for parent functions so that students can play with the slider to see the movement for $kf(x)$ and $f(kx)$. The three parent functions that Algebra 1 is restricted to could add visual confusion, as these do not offer enough of a distinction for the student. Often in these parent functions, $k(f(x))$ and $f(kx)$ are the same. The lessons were written so that the effect taught does apply to all function types that a student will learn throughout high school mathematics courses.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
13.7.1 Warm-Up provides students with an image of dice of multiple colors and structures. Students will be asked to determine the things they notice about the picture and things that they wonder.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
13.7.2 Exploration allows an opportunity for students to explore transformations by a factor with a partner or small group. During this process, students will justify results by explaining methods and processes as well as constructing possible arguments based on evidence.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider the use of digital representations for students to explore as part of this lesson. For example, use Desmos with sliders for students to visualize transformations.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

13.7.1 Warm-Up: Wonder and Notice

Component Overview: *Students will view the provided image to identify what they notice, what they wonder, and a question they could ask about a 5th-grade student.*

13.7.1 Warm-Up: Wonder and Notice

A picture of dice is shown.

1. What do you notice?
   Answers will vary. I notice that this image is a colorful pattern. I notice that this has cubes and expands.

2. What do you wonder?
   Answers will vary. I wonder how many dice there are. I wonder why the colors are in a certain order.

3. Write a question about the picture that you could ask a 5th grade student.
   Answers will vary. How many dice can you estimate to be in the picture? How many dice would be needed to make another row?

This activity provides students with an image of dice of multiple colors and structures. Students will be asked to determine the things they notice about the picture and things that they wonder.

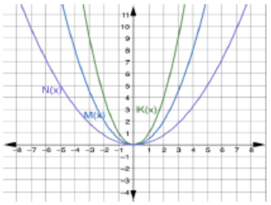
Allow students to share out their questions from part 3 and call on other students to answer the questions from other students.

13.7.2 Exploration: Transformations by a Factor

Component Overview: *Students will engage in an Exploration activity where they will explore transformations by a factor. For this activity, students will need a partner to complete.*

13.7.2 Exploration: Transformations by a Factor

1. The functions $K(x)$, $M(x)$, and $N(x)$ are shown on the graph where $K(x) = x^2$, $M(x) = \frac{1}{2}x^2$, and $N(x) = \frac{1}{5}x^2$. The table of values for each function are also shown.



$y = K(x)$		$y = M(x)$		$y = N(x)$	
x	y	x	y	x	y
-3	9	-3	$\frac{9}{2}$	-3	$\frac{9}{5}$
-2	4	-2	2	-2	$\frac{4}{5}$
-1	1	-1	$\frac{1}{2}$	-1	$\frac{1}{5}$
0	0	0	0	0	0
1	1	1	$\frac{1}{2}$	1	$\frac{1}{5}$
2	4	2	2	2	$\frac{4}{5}$
3	9	3	$\frac{9}{2}$	3	$\frac{9}{5}$

Discuss with your partner the effect the coefficient has on the y-values. Then, complete the summary.

The coefficient of $M(x) = \frac{1}{2}x^2$ is $\frac{1}{2}$. The y-values of $M(x)$ can be found by multiplying the y-values of $K(x)$ by $\frac{1}{2}$ for each corresponding x-value. The coefficient of $N(x) = \frac{1}{5}x^2$ is $\frac{1}{5}$. The y-values of $N(x)$ can be found by multiplying the y-values of $K(x)$ by $\frac{1}{5}$ for each corresponding x-value.

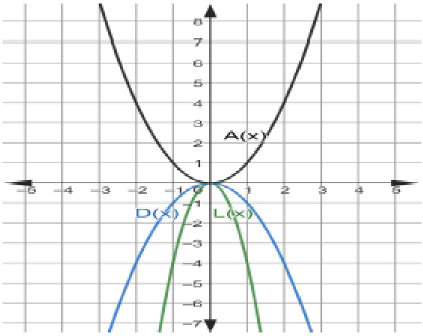
Consider enhancing this by adding a digital exploration component using Desmos. These graphs have sliders where students can manipulate them and see the shifting of the graph happen in response to their values.

Consider leveraging a Think-Pair-Share routine for the first question. Allow students a few minutes to study the graph, function, and tables to give some processing time. Then have students work with their partner to complete the remainder of this activity.

Some students may be overwhelmed by the text in this question. Consider providing a word bank with the values that will be needed to fill in the blank. If you do this, add values that would be incorrect so students have to use their knowledge, not process-of-elimination strategies.

13.7.2 Exploration: Transformations by a Factor (continued)

2. The functions $A(x)$, $D(x)$, and $L(x)$ are shown on the graph, where
   $A(x) = x^2$,
   $D(x) = -x^2$, and
   $L(x) = -4x^2$. The table of values for each function are also shown.



$y = A(x)$		$y = D(x)$		$y = L(x)$	
x	y	x	y	x	y
-3	9	-3	-9	-3	-36
-2	4	-2	-4	-2	-16
-1	1	-1	-1	-1	-4
0	0	0	0	0	0
1	1	1	-1	1	-4
2	4	2	-4	2	-16
3	9	3	-9	3	-36

Discuss with your partner the effect the coefficient has on the y-values. Then, complete the summary.

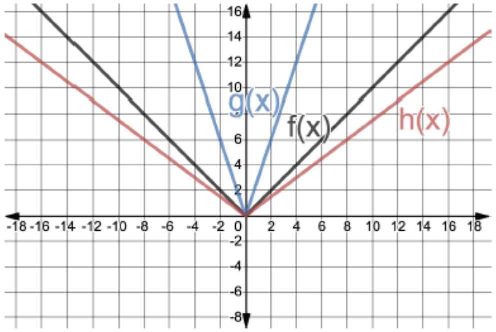
The coefficient of $D(x) = -x^2$ is $\underline{-1}$. The y-values of $D(x)$ can be found by multiplying the y-values of $A(x)$ by $\underline{-1}$ for each corresponding x-value. The coefficient of $L(x) = -4x^2$ is $\underline{-4}$. The y-values of $L(x)$ can be found by multiplying the y-values of $A(x)$ by $\underline{-4}$ for each corresponding x-value.

Ensure students are getting individual processing time to review the information about the function prior to working with their partner to allow for additional processing time that may be necessary.

Some students may be overwhelmed by the text in this question. Consider providing a word bank with the values that will be needed to fill in the blank. If you do this, add values that would be incorrect so students have to use their knowledge, not process-of-elimination strategies.

13.7.2 Exploration: Transformations by a Factor (continued)

3. The functions $f(x)$, $g(x)$, and $h(x)$ are shown on the graph where $f(x) = |x|$, $g(x) = |3x|$, and $h(x) = |0.75x|$. The table of values for each function are also shown.



$y = f(x)$		$y = g(x)$		$y = h(x)$	
x	y	x	y	x	y
-3	3	-1	3	-4	3
-2	2	$-\frac{2}{3}$	2	$-2\frac{2}{3}$	2
-1	1	$-\frac{1}{3}$	1	$-\frac{4}{3}$	1
0	0	0	0	0	0
1	1	$\frac{1}{3}$	1	$\frac{4}{3}$	1
2	2	$\frac{2}{3}$	2	$2\frac{2}{3}$	2
3	3	1	3	4	3

Discuss with your partner the effect the coefficient has on the table of values. Then, complete the table.

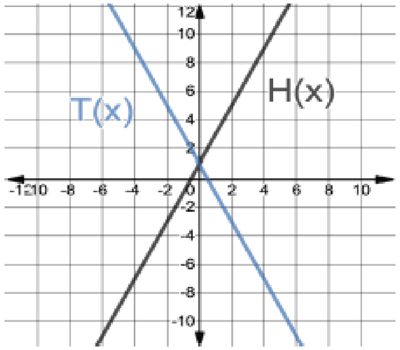
Function	Values that Changed from $f(x) = \lvert x \rvert$	Description of the Change
$g(x) = \lvert 3x \rvert$	○ x-values ○ y-values	○ Divided by 3 ○ Multiplied by _____
$h(x) = \lvert 0.75x \rvert$	○ x-values ○ y-values	○ Divided by 0.75 ○ Multiplied by _____

When circulating during this activity, consider asking questions about the functions in each question. This can include the transformation type, the intercept value, or other features students should know to keep a spiraled review built into this process.

13.7.2 Exploration: Transformations by a Factor (continued)

4. The functions $H(x)$ and $T(x)$ are shown on the graph where $T(x) = H(-x)$. The table of values for each function are also shown.

$y = H(x)$		$y = T(x)$	
x	y	x	y
-3	-5	-3	7
-2	-3	-2	5
-1	-1	-1	3
0	1	0	1
1	3	1	-1
2	5	2	-3
3	7	3	-5



a. Discuss with your partner the effect on the table of values when the x in $H(x)$ was replaced with $-x$ to create $T(x)$.
The y-values of $T(x)$ correspond to the y-values of the opposite x-values of $H(x)$.

b. Discuss with your partner the effect on the graph when the x in $H(x)$ was replaced with $-x$ to create $T(x)$.
It was reflected over the y-axis.

Watch for how students respond to the table. There may be student misconception on what is occurring during each of the transformations.

13.7.3 Bring It Together: The Effect of Transformations of Factors

Component Overview: *Students take what they learned in the 13.7.2 Exploration activity and bring it together to explore the effect of transformations of factors.*

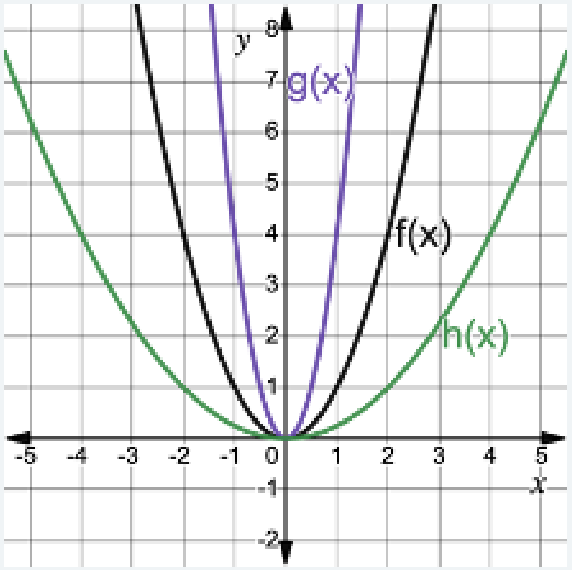
13.7.3 Bring it Together: The Effect of Transformations of Factors

1. Complete the table by determining which values in the table of values of $f(x)$ change, then describe the effect of the factor.

Function Notation	k-values	Which Values Change in the Table of Values?	Effect of the Factor
$f(kx)$	$k < -1$	○ x-values ○ y-values	*Divide x-values by the k-value*
$f(kx)$	$-1 < k < 1$, $k \neq 0$	○ x-values ○ y-values	*Divide the x-values by the k-value*
$f(kx)$	$k > 1$	○ x-values ○ y-values	*Divide x-values by the k-value*
$kf(x)$	$k < -1$	○ x-values ○ y-values	*Multiply the y-values by the k-value*
$kf(x)$	$-1 < k < 1$	○ x-values ○ y-values	*Multiply the y-values by the k-value*
$kf(x)$	$k > 1$	○ x-values ○ y-values	*Multiply the y-values by the k-value*

This activity can be converted to a gallery walk or group assignment. Each part of the table can be posted somewhere in the room, and each group can circulate to complete a different component of the table, making notes and observations about the other groups' work.

Teachers may also want to discuss the effect of the factor (horizontal or vertical dilations) in terms of the graph, using caution with language such as stretch/compression. A vertical shrink could also be considered a horizontal stretch and vice versa. For example, g(x) could be described as a vertical stretch, g(x)=4f(x), or a horizontal shrink, g(x)=f(2x). Likewise, h(x) could be described as a horizontal stretch, h(x)=f(0.5x), or a vertical shrink, h(x)=0.25f(x).



13.7.4 Guided Practice: Describing Transformations

Component Overview: Students will complete two practice questions with transformations by a factor given a table of values and a graph, separately.

13.7.4 Guided Practice: Describing Transformations

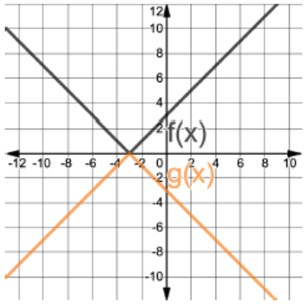
1. The table of values of the function, $y = W(x)$, is shown.

x	-9	-6	-3	0	3	6	9
W	-12	-10	-8	-6	-4	-2	0

Complete the sentence to describe the effect on the x-values for the function, $y = W(4x)$.

The x-values for $W(x)$ will be ○ divided ○ multiplied by 4.

2. The functions $f(x)$ and $g(x)$ are shown on the graph, where $g(x) = -f(x)$.



Complete the sentence to describe the effect on the y-values for the function $y = f(x)$.

The y-values for $f(x)$ will be ○ divided ○ multiplied by -1.

Pay close attention to which question(s) students get incorrect. By missing question 1, but getting question 2, students likely understand the concept when a graph is provided, but haven't connected it to the values in a table or they may have general difficulties with interpreting tables of values.

What would happen to the graph if it were be divided by -1 instead of multiplied?

13.7.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students will complete a Ticket Out the Door that assesses their understanding of transformations with a factor.*

13.7.5 Wrap-Up: Ticket Out the Door

1. Determine how the values in a table of values for $K(x)$ and $N(x)$ will change given that $G(x) = x^2$. Complete the table.

Function	Values that Changed from $G(x) = x^2$	Description of the Change
$K(x) = (3x)^2$	○ x-values ○ y-values	○ Divided by $\underline{3}$ ○ Multiplied by _____
$N(x) = \frac{2}{3}x^2$	○ x-values ○ y-values	○ Divided by _____ ○ Multiplied by $\frac{2}{3}$

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.